Answer Key - Mathematics - Undergraduate Level Assessment 11 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** B
   Options: A) True, True; B) False, False; C) True, False; D) False, True
   Note: Correct option: B - False, False

2. **Answer:** A
   Options: A) 0; B) 1; C) 4; D) 6
   Note: Correct option: A - 0

3. **Answer:** A
   Options: A) -19; B) -10; C) 19; D) 10
   Note: Correct option: A - -19

4. **Answer:** B
   Options: A) 0; B) 3; C) 12; D) 30
   Note: Correct option: B - 3

5. **Answer:** D
   Options: A) True, True; B) False, False; C) True, False; D) False, True
   Note: Correct option: D - False, True

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '3'.

7. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '4 x - 9 y'.

8. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

9. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'False, False'.

10. **Answer:** False
    Options: A) True; B) False
    Note: LLM-generated false statement. Correct answer: 'False, False'.

Long Form Response

11. **Answer:** Let C be the intersection of the horizontal line through A and the vertical line through B. In right triangle ABC, we have $BC = 3$ and $AB = 5$, so $AC = 4$. Let x be the radius of the third circle, and D be the center. Let E and F be the points of intersection of the horizontal line through D with the vertical lines through B and A, respectively, as shown. [asy] unitsize(0.7 cm); draw((-3,0)–(7.5,0)); draw(Circle((-1,1),1),linewidth(0.7)); draw(Circle((3,4),4),linewidth(0.7)); draw(Circle((0.33,0.44),0.44),linewidth(0 dot((-1,1)); dot((3,4)); draw((-1,1)–(-2,1)); draw((3,4)–(7,4)); label("{ A}",(-1,1),N); label("{ B}",(3,4),N); label("{ 1}",(-1.5,1),N); label("{ 4}",(5,4),N); draw((3,4)–(-1,1)–(3,1)–cycle); draw((3,0.44)–(-1,0.44)); draw((-1,1)–(-1,0)); draw((3,1)–(3,0)); draw((-1,1)–(0.33,0.44)); draw((0.33,0.44)–(3,4),dashed); dot((3,1)); dot((3,0.44)); dot((-1,0.44)); dot((0.33,0.44)); label("{ C}",(3,1),E); label("{ E}",(3,0.44),E); label("{ D}",(0.33,0.44),S); label("{ F}",(-1,0.44),W); [/asy] In $\triangle BED$ we have $BD = 4 + x$ and $BE = 4 - x$, so $DE^2 = (4+x)^2-(4-x)^2 = 16x, and$DE $= 4\sqrt{x}. In\triangle$ ADF$we have$AD $= 1+$xandAF=1-x,$so FD^2 = (1+x)^2-(1-x)^2 = 4x, and$FD $= 2\sqrt{x}. Hence, 4 = AC = FD + DE = 2\sqrt{x}+4\sqrt{x} = 6\sqrt{x} and \sqrt{x}$=$\frac{2}{3}, which implies$x=$\{\frac{4}{9}\}$.\$
Note: Students should show all steps in their mathematical solution.

12. **Answer:** 1/2. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key